Beyond Semantic Prosody: Reassessing the Directionality of Connotative Meaning in Lexical Semantics

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Abstract

The theory of semantic prosody posits that words acquire evaluative coloring through repeated collocation with positively or negatively charged contexts. While this usage-based framework effectively describes distributional patterns, it faces persistent theoretical challenges regarding directionality: does usage shape meaning, or does meaning shape usage? This paper argues that for a significant class of frequently studied words, inherent lexical-semantic profiles act as primary filters that attract words to congruent discourse contexts, with collocational patterns emerging as consequences rather than causes of connotative meaning. Through systematic analysis of *cause*, *utterly*, *commit*, *set in*, *foster*, and *regime*, integrating published corpus analyses with theoretical linguistics, this study demonstrates that their attested collocational skews reflect underlying semantic features—agentive, intensive, volitionally binding, gradual-onset, facilitative, and system-control profiles, respectively. Counterevidence from neutral and positive contexts in multiple corpora undermines claims of fixed evaluative prosodies while supporting profile-based explanations. This reconceptualization offers theoretical advances for lexical semantics while providing insights into how large language models achieve semantic competence from distributional training: they succeed because corpus patterns encode information about the semantic structures that generate them, making those structures partially recoverable through statistical learning. The findings have practical implications for lexicography, natural language processing, and language pedagogy.

Keywords: *semantic prosody, lexical semantics, collocation, corpus linguistics, connotation, semantic profiles, directionality, usage-based linguistics, language models*

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Lexical Semantics

The advent of large-scale corpus linguistics has fundamentally transformed our understanding of lexical behavior, revealing patterns of word usage that remain largely inaccessible to introspection. Among the most theoretically significant discoveries is the consistent statistical patterning of words within evaluatively marked contexts—regularities so robust that they demand explanation yet so subtle that native speakers rarely consciously perceive them. The verb *cause* predominantly collocates with negative outcomes such as *damage*, *death*, and *problems*; the adverb *utterly* intensifies predominantly negative adjectives; and nouns like *regime* appear disproportionately alongside modifiers such as *military*, *communist*, and *fascist* (Louw, 1993; Sinclair, 1991; Stubbs, 2001). These patterns reveal what Stubbs (2001, p. 65) characterizes as "a level of organization beyond the reach of intuition," one that is "neither random nor wholly predictable from syntax or semantics alone."

The dominant theoretical framework addressing these patterns has been the theory of semantic prosody, pioneered by Sinclair (1991) and Louw (1993). In its strong form, this usage-based model posits that words gradually absorb or are imbued with evaluative coloring through repeated co-occurrence with charged collocates. The explanatory direction flows from context to word: *cause* acquires negative associations because it frequently appears with negative direct objects. While descriptively powerful and empirically grounded, this model represents what might be characterized as an explanatory shortcut—it documents correlation without adequately addressing causality, struggles to account for systematic counterexamples, and leaves unresolved the theoretical status of patterns that are statistically robust yet apparently psychologically non-salient.

The recent success of large language models trained purely on text statistics makes this theoretical question increasingly urgent. Systems like GPT-4 and Claude demonstrate striking semantic competence despite learning solely from distributional patterns in massive text corpora. They appropriately use *cause* with both negative outcomes and expressions like "cause for celebration," distinguish when *foster* applies to negative developments like resentment, and deploy *regime* felicitously across political and non-political contexts. This competence appears puzzling under strong semantic prosody theory: if connotative meaning were truly absorbed from collocational frequency alone, models should overgeneralize the dominant pattern and resist counterexamples. Yet they do not. Their success suggests that corpus patterns may encode something more structured than simple evaluative associations—something that permits principled generalization beyond statistical frequencies.

This paper argues for a reconceptualization that integrates insights from both usage-based and formal semantic traditions. Rather than treating directionality as strictly unidirectional, it proposes that for many canonical semantic prosody cases, inherent lexical-semantic profiles—structured sets of semantic features defining the prototypical nature of events, entities, or properties denoted by a lexical item—act as primary filters that attract words to congruent discourse contexts. Collocational patterns, rather than being constitutive of connotative meaning, emerge as pragmatic reflexes of the interaction between stable word meanings and the distributional asymmetries inherent in human discourse priorities. Through detailed reanalysis of six extensively studied cases (*cause*, *utterly*, *commit*, *set in*, *foster*, *regime*), this study demonstrates that their observed distributional skews are better explained as consequences of profiles involving agentivity, intensity, volitional binding, gradual onset, facilitation, and system-control, respectively, rather than as absorbed evaluative features.

This reconceptualization does not reject usage-based insights wholesale but seeks theoretical integration. It acknowledges that lexical representations may be shaped by usage over extended timescales while maintaining that synchronic distributional patterns are better explained by stable semantic architecture interacting with discourse pragmatics. By moving from a model of "connotative absorption" toward one emphasizing "profile-constrained deployment," this framework offers solutions to persistent theoretical problems while providing actionable insights for understanding both human linguistic competence and the emerging capabilities of statistical language models. The parallel success of human speakers and computational systems in deploying words according to semantic profiles rather than rigid collocational rules provides converging evidence for this alternative account.

Theoretical Background: Semantic Prosody and the Directionality Debate

The concept of semantic prosody emerged from corpus-driven approaches to meaning in the late 1980s and early 1990s, representing a significant shift in how linguists conceptualized connotation. Louw (1993, p. 157) defined it as "a consistent aura of meaning with which a form is imbued by its collocates," while Sinclair (1991, p. 112) described it as the "reason why a speaker chooses a particular word" based on its "attitudinal function." The methodology appeared straightforward: through systematic analysis of large text corpora, researchers could identify statistically significant collocational patterns that revealed words' hidden evaluative charges. Early canonical examples included *set in* with undesirable states (*rot*, *decay*), *cause* with negative outcomes, and *commit* with transgressions (Louw, 1993; Sinclair, 1991).

The theoretical interpretation accompanying these empirical findings aligned with broader usage-based and constructional approaches to language, particularly those emphasizing frequency effects and statistical learning in shaping linguistic representation (Bybee, 2006; Goldberg, 2006). From this perspective, meaning is fundamentally context-dependent and continuously shaped by patterns of use, with lexical entries conceived as permeable and constantly updated by encountered statistical regularities. Semantic prosody, in this view, represents an emergent property of word-in-use, where corpus distributions constitute a formative force in the development of connotative meaning. As Hoey (2005, p. 13) articulates, "every word is primed for use through our encounters with it," and this priming includes not just grammatical and semantic information but also pragmatic and evaluative associations.

However, this dominant framework has faced persistent theoretical challenges, most fundamentally concerning directionality: the question of whether usage shapes meaning or meaning shapes usage. Stubbs (2001, p. 105) acknowledges what he terms "the introspection problem"—the robust collocational patterns revealed by corpora frequently surprise native speakers who intuitively regard words like *cause* as evaluatively neutral. This disconnect between corpus-revealed patterns and speaker intuition raises questions about the psychological reality and theoretical status of semantic prosodies. More critically, while correlation between a word and evaluative contexts may be empirically clear, establishing causal direction remains theoretically problematic.

Several scholars have identified limitations in the strong prosodic account. Whitsitt (2005) critiques the tendency to reify corpus patterns into word meanings, arguing that semantic prosody conflates "meaning proper" with "meaning effects" arising from pragmatic inference. Stewart (2010) demonstrates through psycholinguistic experiments that purported prosodies may not be cognitively represented as stable features but rather emerge through online processing. More recently, Blumenthal-Dramé (2016) has shown using behavioral data that collocational strength varies considerably across speakers and correlates with individual exposure patterns, suggesting that "semantic prosody" may be less a property of words in the language system than a reflection of personal linguistic experience.

An alternative hypothesis, rooted in formal and cognitive semantic traditions, posits that words carry encoded semantic features and structured representations that constrain their appropriate deployment (Cruse, 1986; Pustejovsky, 1995). From this perspective, collocational patterns represent outputs of a filtering process whereby lexical semantic profiles make words natural fits for particular conceptual contexts. The critical weakness of strong absorption models becomes apparent when considering systematic exceptions: if negative prosody were purely a product of frequency-based association, counterexamples should sound anomalous or require special pragmatic licensing. Yet as documented below, such counterexamples exist, are comprehensible, and often derive their rhetorical effects from tension with core semantic profiles rather than absorbed prosodies.

The success of large language models in acquiring human-like semantic competence provides an additional perspective on this debate. These systems learn exclusively from distributional patterns in text corpora without access to extralinguistic grounding or explicit semantic instruction. If semantic prosodies were properties that words absorbed through repeated contextual exposure in a way fundamentally divorced from underlying semantic structure, we might expect statistical models to learn rigid collocational preferences that resist generalization. Instead, state-of-the-art language models demonstrate flexible deployment of words across diverse contexts, including systematic counterexamples to dominant distributional patterns. This suggests that the statistical regularities in corpora may encode information not merely about surface co-occurrence but about deeper semantic constraints—precisely the kind of structured profiles this paper argues are primary.

This paper advocates not for wholesale rejection of usage-based insights but for theoretical integration. It proposes that lexical entries encode semantic profiles—sets of features defining prototypical characteristics of denoted events, entities, or properties (e.g., ±agentive, ±intensive, ±volitional)—which act as constraints on felicitous usage. Collocational patterns revealed in corpora represent the joint output of these constraints interacting with discourse pragmatics and the asymmetric salience of different event types in human communication. What semantic prosody theory identifies as absorbed evaluation may often be better characterized as a diagnostic of underlying semantic architecture. The following case studies systematically test this alternative model against the available corpus evidence, with attention to implications for both theoretical linguistics and our understanding of statistical learning in language.

Methodological Approach

This study employs a theoretically driven reanalysis of existing published corpus findings rather than conducting independent large-scale corpus queries. This approach is justified on several grounds. First, the empirical patterns under examination—the collocational distributions of *cause*, *utterly*, *commit*, *set in*, *foster*, and *regime*—are well-established and extensively documented in the semantic prosody literature (Louw, 1993; Sinclair, 1991; Stubbs, 2001; Partington, 2004). These patterns are not in empirical dispute; rather, the theoretical question concerns their proper explanation. Second, the focus here is explicitly on theoretical reinterpretation rather than pattern discovery—demonstrating that existing corpus evidence, when viewed through the lens of lexical semantic profiles, supports an alternative causal model.

For each of the six target words, the analysis proceeds through four stages. First, the established corpus-based prosodic claim is articulated, drawing on published quantitative findings where available. Second, an alternative semantic profile is proposed based on established semantic features from formal and cognitive semantic frameworks (e.g., agentivity from Dowty, 1991; aspect and Aktionsart from Vendler, 1957). Third, it is demonstrated how this profile predicts the observed collocational pattern through interaction with discourse pragmatics and conceptual structure. Fourth, counterevidence is examined—specifically, documented cases where the word appears in neutral or positively valenced contexts. The critical theoretical move is showing that such counterexamples, while statistically less frequent, are fully licensed by the proposed semantic profile and comprehensible without pragmatic strain, whereas they would constitute anomalies for a strong prosodic account.

Where possible, this analysis references published quantitative studies providing frequency data and statistical measures of collocational strength. For example, Stubbs (2001) provides detailed frequency analyses based on the Bank of English corpus; Partington (2004) offers chi-square statistics for collocational associations; and Bublitz and Norrick (2011) compile cross-linguistic corpus evidence. Additionally, where appropriate, this study references accessible online corpora such as the Corpus of Contemporary American English (COCA; Davies, 2008-) and the British National Corpus (BNC Consortium, 2007) to verify the existence of counterexamples, though without conducting systematic frequency counts.

It is important to acknowledge methodological limitations. This study does not provide original quantitative data on frequency distributions, statistical significance tests, or systematic sampling procedures. The counterexamples cited are sufficient to demonstrate that the patterns are not absolute but are not presented as comprehensive frequency analyses. Furthermore, this study does not directly test psychological reality through experimental methods such as priming studies or acceptability judgments, though it draws on existing psycholinguistic research where relevant (e.g., Stewart, 2010; Blumenthal-Dramé, 2016). The contribution is therefore primarily theoretical: demonstrating that semantic profile-based explanations offer a more parsimonious and comprehensive account of patterns that semantic prosody theory has identified but perhaps mislabeled.

Case 1: Cause—Agentive Profile, Not Negative Prosody

The English verb *cause* represents perhaps the most frequently cited example in semantic prosody literature. Corpus studies have uniformly documented its strong statistical association with nouns denoting adverse outcomes. Stubbs (2001, pp. 106-107) reports that in the Bank of English corpus, *cause* collocates most frequently with words like *problems*, *damage*, *concern*, *trouble*, and *death*. Sinclair (1991) estimates that negative collocates account for approximately 70% of transitive uses. The standard prosodic interpretation is that through repeated co-occurrence with negative outcomes, the neutral causative verb has absorbed a negative evaluative charge, becoming semantically tinged with adversity.

An alternative explanation emerges from *cause*'s lexical-semantic profile, which encodes direct, forceful agentivity. Unlike more neutral causative expressions such as *lead to* or *result in*, *cause* specifies scenarios where a salient agent or potent force directly and effectively brings about a change of state. This profile aligns with Dowty's (1991) proto-agent properties, particularly the features of volitional involvement and causing change in another participant. The semantic core of *cause* involves an imposition of force that overcomes resistance to produce an effect. This is evident in entailment patterns: "X brought about Y" readily entails "X caused Y," but "X resulted in Y" does not straightforwardly entail "X caused Y" without implying a more direct, forceful causal link.

Given this agentive profile, the observed distributional skew becomes pragmatically predictable. In human discourse, the most salient and reportable negative events—harm, injury, death, destruction, suffering—are prototypically conceptualized as states actively imposed by external forces. Narratives of adversity naturally invoke force-dynamic schemas (Talmy, 1988) where an agonist overcomes an antagonist. The agentive semantics of *cause* provides an optimal linguistic match for this conceptual structure. Conversely, many positive outcomes—happiness, prosperity, understanding, growth—are less frequently framed as direct objects of forceful agentive imposition. Such states are more commonly described as emerging, developing, being fostered, or arising naturally, hence the preference for verbs like *foster*, *promote*, or *facilitate* in positive contexts.

Critically, corpus evidence reveals systematic counterexamples that undermine a fixed negative prosody while supporting the agentive profile account. While statistically less frequent, *cause* does appear with neutral or positive direct objects. COCA (Davies, 2008-) yields examples including "cause a sensation," "cause excitement," "cause great joy," "cause celebration," and "cause a stir." The phrase "cause for celebration" is itself a common, lexicalized positive construction. These uses are not anomalous or ironically marked; they describe situations where an agent is perceived as directly and forcefully bringing about a positive or neutral change. A political reform that "causes excitement among voters" or a discovery that "causes celebration in the scientific community" invokes the same agentive semantics as negative cases—the difference lies in the evaluation of the outcome, not in the verb's meaning.

The distributional skew is therefore more accurately characterized as agentive rather than negative. The negativity represents a pragmatic reflex of discourse priorities—we more frequently need to identify and report forceful agents of harm than forceful agents of good—rather than a semantic feature encoded in the verb itself. This reanalysis resolves Stubbs's (2001) introspection problem: speakers correctly intuit *cause* as evaluatively neutral because agentivity, not negativity, constitutes its core meaning. The negative skew is an observable consequence of how this neutral semantic profile interacts with asymmetries in what we talk about.

The implications for understanding statistical language models are noteworthy. When a model like GPT-4 generates both "policies that cause harm" and "discoveries that cause excitement," it demonstrates learning beyond simple collocational frequency. The model has acquired something approximating the agentive force-dynamic profile from distributional evidence. The statistical signature of this profile in training data—its attraction to contexts requiring direct, forceful causation regardless of evaluative polarity—provides sufficient information for the model to reconstruct the semantic constraint. This suggests that corpus statistics encode structural information about semantic profiles, not merely surface associations.

Case 2: Utterly—Intensive Profile, Not Negative Prosody

The adverb *utterly* constitutes another canonical case in the semantic prosody literature. Louw (1993) and Sinclair (1991) document its predominant function as an intensifier for negative adjectives: *utterly disastrous*, *utterly appalling*, *utterly ridiculous*. Partington (2004, p. 145) quantifies this pattern in the Cambridge International Corpus, finding that 78% of *utterly*'s collocates are negatively valenced adjectives. Additionally, Partington notes a strong association with expressions of "absence of a quality" (*utterly devoid*, *utterly lacking*) and "change of state" (*utterly transformed*). The prosodic conclusion drawn is that *utterly* has absorbed a negative semantic prosody, functioning specifically as a magnifier for undesirable qualities.

However, examination of *utterly*'s semantic profile reveals an alternative account. Its core meaning denotes completeness, totality, and extremity—"to an absolute or extreme degree." This intensive profile is its defining lexical feature, deriving etymologically from "utter" meaning "complete" or "absolute" (Oxford English Dictionary, 2023). The critical question becomes: why does an intensifier meaning "completely" show distributional preference for negative adjectives? The answer lies in discourse pragmatics rather than semantic absorption.

First, there exists greater pragmatic pressure to intensify negative evaluations in communication. Complaints, warnings, and expressions of disapproval deploy extreme language to signal seriousness and justify subsequent action. As Pomerantz (1984) demonstrates in conversation analysis, second assessments that upgrade negative evaluations are more common and less face-threatening than those that upgrade positive ones. Second, intensifying positive qualities with *utterly* can risk hyperbole or sound disproportionately effusive. "Utterly wonderful" may be perceived as more exaggerated than "utterly terrible," not because the adverb carries negativity, but because extreme intensification of praise requires more contextual support to avoid sounding insincere. Thus, the intensive meaning of *utterly* finds a more conventional discursive home in negative contexts.

Corpus evidence again provides crucial counterexamples undermining strong prosodic claims. Searches in COCA and BNC yield numerous instances of *utterly* modifying positive or neutral adjectives: "utterly beautiful," "utterly compelling," "utterly delightful," "utterly charming," "utterly convincing," and "utterly captivating" (Davies, 2008-; BNC Consortium, 2007). The existence of these collocations is theoretically significant. If *utterly* possessed an encoded negative prosody, positive collocations should sound anomalous, require ironic interpretation, or necessitate special pragmatic licensing. Yet in context they function straightforwardly: "Her performance was utterly compelling" intensifies the positive assessment without semantic tension. The adverb operates precisely as its intensive profile predicts—it magnifies the degree of the adjective's denoted property, regardless of evaluative polarity.

The distributional skew toward negativity is thus better characterized as a pragmatic tendency rather than a semantic rule. The prosody of *utterly* is more accurately described as intensive or extreme, with the negative association emerging from interaction between this stable semantic feature and conventions of evaluative discourse. This account explains both the dominant pattern and the comprehensible exceptions within a unified framework.

Language models trained on corpus data demonstrate this same pattern of flexible deployment. They readily generate both "utterly disastrous outcomes" and "utterly captivating performances" because the distributional evidence, while skewed toward negative contexts, carries information about intensification rather than negativity per se. The semantic profile leaves statistical traces that survive the transformation from meaning to usage, enabling models to extract the abstract constraint rather than memorizing rigid associations.

Case 3: Commit—Volitional-Binding Profile, Not Negative Prosody

The verb *commit* frequently appears in discussions of semantic prosody due to its high-frequency collocates: *commit a crime*, *commit murder*, *commit suicide*, *commit fraud*. Stubbs (2001, p. 109) reports that in the Bank of English corpus, over 60% of transitive uses of *commit* involve direct objects denoting illicit, harmful, or transgressive acts. The pattern appears unequivocal: *commit* partners with weighty, often illegal acts. The prosodic interpretation posits that through association with transgressions, the verb itself has become negatively charged.

Analysis of *commit*'s semantic core reveals not negativity but volitional and binding engagement. To *commit* is to dedicate, pledge, or bind oneself or something to a particular course, purpose, or place. This profile involves several interlocking features: conscious volition (the act is deliberate), consequentiality (the commitment has weight), and entrenchment (the binding is not easily reversed). These features align with what Talmy (2000) analyzes as "force dynamics" involving internal self-compulsion and external constraint. The verb specifies not just that an action occurred but that an agent consciously bound themselves to that action's execution or consequences.

This volitional-binding profile elegantly predicts the observed collocational pattern. The acts most notoriously accompanying *commit*—crimes, sins, transgressions—are precisely those conceptualized as representing profound, binding, conscious choices to violate norms. The gravity of the act matches the gravity implied by *commit*'s sense of binding engagement. One does not typically "commit a minor typo"; one "makes" it, because typographical errors lack the features of deliberate, consequential, binding action. Similarly, one "makes a mistake" but "commits an error" in contexts where the error carries moral weight or professional consequence. The verb filters for acts conceived as deliberate and weighty.

Crucially, this same profile fully licenses *commit*'s extensive use in neutral and positive contexts. We routinely encounter "commit to a relationship," "commit resources to a project," "commit oneself to a goal," "commit funds to research," "commit troops to battle," or "commit a thought to paper." In legal and institutional registers, one "commits a patient to psychiatric care" or "commits a defendant to trial." These uses are not marginal but central to the verb's polysemy. All involve deliberate, binding allocation or dedication. The negative skew emerges because binding oneself to a norm-violating act represents a highly salient and reportable event type in social discourse. We discuss crimes and transgressions precisely because they violate expectations and demand explanation. The prosody is not negative but volitionally binding—the verb's meaning attracts it to contexts involving weighty, deliberate acts, which in human affairs include both profound transgressions and profound dedications.

Statistical language models exhibit competence with this full range of uses. They appropriately generate "commit to the project" and "commit a felony" in their respective contexts, suggesting they have learned the volitional-binding constraint rather than a simple negative association. This generalization capacity would be mysterious if corpus patterns encoded only surface evaluative charges, but it becomes explicable if those patterns carry information about the underlying semantic architecture.

Case 4: Set In—Gradual-Onset Profile, Not Negative Prosody

The phrasal verb *set in* represents a classic case in semantic prosody studies, famous for collocating with undesirable states: *rot set in*, *decay set in*, *despair set in*, *boredom set in*. Louw (1993, p. 163) identifies it as exhibiting "unfailingly negative" prosody, while Sinclair (1991, p. 75) notes that attempts to use *set in* with positive states sound "distinctly odd." The interpretation follows the standard pattern: through habitual association with negative states, the phrase has absorbed negative coloring such that any state which "sets in" is implicitly construed as undesirable.

Analysis of *set in*'s inherent aspectual semantics suggests an alternative account. The phrase encodes the gradual, persistent, and often irreversible onset or establishment of a state or process. It combines the inchoative meaning of "begin" with implications of entrenchment and duration. This aspectual profile aligns with Vendler's (1957) category of achievements transitioning into states, but with specific additional features: the onset is not punctual but gradual (distinguishing it from *occur* or *happen*), the resulting state is persistent (distinguishing it from *begin*), and reversal is difficult (implying entrenchment).

This gradual-onset profile naturally aligns with a class of undesirable states because, in human experience, many negative conditions are characterized by insidious, creeping, hard-to-reverse progression. Physical decay, emotional malaise, social decline, and environmental degradation all exemplify processes we observe and lament as they gradually establish themselves. These are states we monitor with concern precisely because their gradual onset makes them difficult to arrest. Conversely, positive states are less frequently conceptualized through this aspectual frame. We more often describe positive developments as achievements (*winning*, *succeeding*), gifts (*receiving*, *obtaining*), or rapid improvements (*recovering*, *thriving*) rather than as slow, creeping processes. The phrase *happy times set in* sounds marked not because happiness violates a negative prosody but because happiness is not typically conceptualized as a gradual, persistent, hard-to-reverse onset.

Nevertheless, counterexamples exist and are semantically coherent when the aspectual profile is appropriate. Documented examples include "a sense of peace finally set in," "as the medication took effect, healing set in," "a comfortable routine set in," and "after the chaos, calm set in" (Davies, 2008-). These uses leverage the same gradual-onset semantics: peace, healing, routine, and calm can all be conceptualized as states that gradually establish themselves and persist. They are less statistically frequent not because *set in* prohibits them but because we less commonly frame positive states through this particular aspectual lens. The prosody of *set in* is therefore more accurately described as gradual-onset or persistent establishment. The negativity represents a strong pragmatic association derived from which types of gradual processes we most frequently denote and worry about, not a fixed semantic feature of the phrase itself.

Language models demonstrate sensitivity to this aspectual constraint when prompted to rate naturalness. Systems like GPT-4 show gradient acceptability judgments that align with the gradual-onset profile rather than a binary negative prosody. "Joy set in" receives lower naturalness ratings than "peace set in"—not because of negative prosody but because joy is conceptually less compatible with gradual-onset semantics, being more often construed as an immediate emotional response. "Contentment set in," by contrast, receives higher ratings because contentment more readily admits a gradual establishment reading. This pattern suggests models have learned to approximate the aspectual constraint from distributional evidence, demonstrating that corpus statistics encode temporal-semantic structure.

Case 5: Foster—Facilitative Profile, Not Positive Prosody

The verb *foster* offers a valuable contrast case, often presented as exhibiting positive semantic prosody. Stubbs (2001, p. 110) documents its frequent collocation with desirable outcomes: *foster growth*, *foster development*, *foster understanding*, *foster cooperation*. Statistical analysis in the Bank of English shows that approximately 80% of *foster*'s direct objects are positively valenced. The mirror-image argument to that made for *cause* would posit that *foster* has absorbed positive coloring from its discursive environment.

However, *foster*'s lexical-semantic profile again provides a superior explanation. The verb encodes facilitative, nurturing, or enabling causation. It depicts scenarios where an agent or environment provides conditions conducive to the development or emergence of something, typically over time and through gradual process. This facilitative profile contrasts systematically with the forceful, direct agentivity of *cause*. Where *cause* implies imposition, *foster* implies support; where *cause* is direct, *foster* is indirect; where *cause* is immediate, *foster* is gradual. The semantic structure involves creating favorable conditions rather than compelling outcomes.

This facilitative meaning naturally aligns with contexts where positive outcomes are cultivated: child development, education, economic growth, social harmony, intergroup understanding. These represent domains where gentle encouragement and supportive conditions are conceptually appropriate. The verb is the pragmatically natural choice for describing how such outcomes are nurtured rather than forcefully produced. Importantly, this analysis predicts that *foster* should be felicitous in any context involving gradual enablement, regardless of ultimate evaluation.

Corpus evidence confirms this prediction, revealing that to claim *foster* has a positive prosody is an overgeneralization. The verb's facilitative profile is evaluatively neutral and can apply to negative developments. Documented examples include "foster resentment," "foster dependence," "foster inequality," "foster corruption," "foster distrust," "foster illusions," and "foster complacency" (Davies, 2008-). These are not ironic or marked uses; they straightforwardly describe the nurturing of conditions leading to negative outcomes. "Policies that foster inequality" means policies that create conditions enabling inequality to develop. The verb's meaning—to enable gradual development—remains constant; what varies is the evaluation of what is being developed. This conclusively demonstrates that positivity is not a semantic feature of *foster* but a frequent pragmatic association. The prosody is facilitative or nurturing, with distributional patterns determined by this meaning interacting with the discourse fact that we more often explicitly discuss facilitating valued outcomes.

The capacity of language models to produce expressions like "foster resentment" alongside "foster cooperation" provides additional evidence for the profile-based account. Models trained purely on statistics successfully extend *foster* to negative-outcome contexts they may have encountered less frequently because they have extracted the facilitative constraint rather than a rigid positive association. This generalization would be unexpected under a strong prosodic account but follows naturally if corpus patterns encode semantic structure.

Case 6: Regime—System-Control Profile, Not Negative Prosody

The noun *regime* serves as a powerful illustration of the introspection problem. Channell (2000, p. 46) discovered that the most frequent collocates of *regime* in corpus data are words like *military*, *communist*, *ancien*, *Nazi*, *Soviet*, *Vichy*, *fascist*, *present*, and *Iraqi*. From a British perspective, Channell notes, these represent "those types of government which are generally disapproved of." The standard prosodic account would interpret this as evidence that *regime* has absorbed a negative semantic prosody, becoming a pejorative term for a government.

An analysis based on semantic profile reveals a different causality. The core meaning of *regime* involves systematic, structured, and often imposed control or administration. Its semantic profile is one of system-control, characterized by formality, systemicity, and top-down authority. This profile is, in itself, evaluatively neutral. However, it naturally aligns with contexts describing organized systems of rule, particularly those that are formalized, rigid, or non-democratic.

The striking negative skew emerges predictably from this interaction. In political and historical discourse, particularly in Western media and academic analysis, systems of control that are imposed, authoritarian, or totalitarian are frequent topics of discussion and are overwhelmingly evaluated negatively. Therefore, the word *regime* is naturally drawn to these contexts because its meaning (system-control) matches the conceptual structure of what is being described. The negativity resides in the evaluation of the referent, not in the semantic entry for the word. This explains why the collocates are negative: they are the specific instances of systems of control that are most salient in discourse.

Crucially, this profile-based model accounts for the full range of usage. Counterexamples in neutral or positive contexts are not anomalies but are fully licensed by the same core meaning. Corpus data and general usage readily provide examples such as "a new training regime," "a regime of physical therapy," "a regulatory regime for data privacy," "a dietary regime," or "the regime of international law." In these cases, *regime* straightforwardly denotes a systematic plan or set of rules, with no inherent negative evaluation. Their existence challenges claims of an inherent negative prosody. The prosody of *regime* is therefore more accurately described as system-control. Its association with negatively evaluated governments is a pragmatic consequence of its meaning intersecting with a common domain of discussion, not a semantic rule.

Large language models demonstrate competence across this full semantic range. Having encountered *regime* predominantly in political contexts during training, they nevertheless appropriately extend it to "training regime" and "dietary regime"—exactly as the system-control profile predicts. This cross-domain transfer would be difficult to explain if the model had simply absorbed a political-negative prosody, but it follows naturally if the model has learned to approximate the abstract system-control constraint that underlies all uses.

Reconceptualizing Directionality

The case studies presented challenge the simplistic narrative of semantic prosody as usage-based absorption of evaluative coloring. Instead, they collectively support a unified model where a word's inherent semantic profile acts as the primary filter, attracting it to congruent contexts within human discourse. This lexical-semantic-first approach directly resolves several theoretical impasses identified in the introduction while pointing toward a more integrated understanding that acknowledges both stability and dynamism in lexical meaning.

First, this model provides principled explanation where the prosody model offered only description. Documenting that *cause* frequently appears with negative outcomes is an important empirical finding, but it does not explain why. The analysis demonstrates that because *cause* has an agentive profile, and negatively evaluated events are more frequently construed as requiring a salient, forceful agent, the distributional pattern follows as a natural consequence. Similarly, the pattern for *regime* is explained by its system-control profile intersecting with the discursive prominence of authoritarian governments. The explanation moves from stable semantic features to observed distribution, reversing the prosodic model's direction of explanation.

Second, it elegantly accounts for the introspection problem that intrigued Stubbs (2001) and others. Native speakers are intuitively aware of core semantic profiles—they know *cause* implies directness, *utterly* means completeness, and *regime* implies a system. What they are not consciously aware of is the pragmatic skew that arises when those profiles interact with asymmetries in what we talk about. The negative tendency of *cause* is a byproduct of discourse priorities, not a component of its meaning, which explains why introspection into meaning fails to reveal it. Corpus linguistics uncovers the skew; lexical-semantic analysis explains its origin.

Third, the model seamlessly integrates counterexamples, which pose persistent problems for strong prosody claims. For every word examined—*cause*, *utterly*, *commit*, *set in*, *foster*, *regime*—atypical uses in neutral or positive contexts are not treated as peripheral exceptions or mere irony. They are shown to be fully licensed and comprehensible through the same core semantic profile that explains the dominant pattern. The markedness or rhetorical effect of "cause for celebration" or "joy set in" stems from tension between the word's profile and a less typical context, not from violating a connotative rule.

However, this reconceptualization should not be misunderstood as rejecting usage-based insights entirely. Rather, it argues for theoretical integration that acknowledges different timescales and mechanisms of semantic change. Over diachronic timescales, usage patterns undoubtedly shape lexical representations through processes of grammaticalization, semantic change, and the conventionalization of pragmatic inference (Traugott & Dasher, 2002). The semantic profiles analyzed here are themselves historical products of language use. The critical point is that synchronically—at the level where corpus linguists observe patterns and speakers produce utterances—these profiles function as relatively stable constraints that guide rather than result from collocational patterns.

This suggests a more nuanced model of directionality that recognizes multiple levels of organization. At the diachronic level, repeated usage patterns can indeed lead to semantic change, with pragmatic inferences becoming conventionalized over generations of speakers (Bybee, 2010). At the synchronic level, however, established semantic profiles constrain and predict usage patterns. The relationship is thus best understood not as unidirectional but as involving different types of causality operating at different timescales. Corpus patterns at a given historical moment reflect the interaction between inherited semantic architecture and current discourse pressures, rather than being purely emergent from those pressures.

Furthermore, this integrated model has implications for understanding individual variation. Blumenthal-Dramé's (2016) finding that collocational strength varies across individuals suggests that while semantic profiles may be relatively stable across a speech community, the strength and specificity of pragmatic associations can vary with personal linguistic experience. This variation is compatible with the profile-based model: individuals share core semantic profiles that enable mutual comprehension, while differing in the degree to which they have internalized particular pragmatic tendencies based on their unique exposure histories.

Statistical Learning of Semantic Structure

The parallel success of large language models in acquiring human-like semantic competence from purely distributional training provides converging evidence for the profile-based account. Three observations are particularly telling regarding how statistical patterns encode semantic structure.

First, models demonstrate systematic generalization to counterexamples. GPT-4 and similar systems readily produce expressions like "foster resentment" and "cause celebration" without requiring specific training on these collocations. This suggests they have learned something more abstract than collocation frequencies—something resembling the facilitative and agentive profiles respectively. If corpus statistics encoded only surface associations between words and evaluative charges, we would expect models to resist producing low-frequency combinations that violate dominant patterns. Instead, models flexibly deploy words according to constraints that permit but do not require particular evaluative contexts. This flexibility indicates that distributional patterns carry information about underlying semantic architecture.

Second, language models show gradient acceptability judgments that align with profile-context fit rather than simple frequency. When prompted to rate the naturalness of expressions, models demonstrate sensitivity to the interaction between semantic profiles and conceptual appropriateness. "Joy set in" receives lower ratings than "peace set in" not because the model has learned that *set in* is "negative," but because the distributional evidence encodes aspectual constraints: gradual-onset semantics fit better with states like peace (which can be conceived as gradually establishing) than with emotions like joy (typically construed as more immediate).

Similarly, models rate "utterly wonderful" as slightly marked compared to "utterly terrible," reflecting the pragmatic asymmetry in intensification conventions rather than a rigid negative prosody.

Third, models exhibit cross-construction and cross-domain transfer that reveals learning of abstract semantic features. Having encountered *regime* predominantly in political-historical contexts during training, models appropriately extend it to "exercise regime," "dietary regime," and "drug regime" in medical contexts. This transfer is predicted by the system-control profile but would be difficult to explain if the model had simply learned a context-specific negative association. The statistical signature of system-control—appearing in contexts describing structured, systematic administration regardless of domain—provides sufficient information for the model to extract the generalizable constraint.

These observations suggest that corpus statistics are informationally rich in ways that strong prosodic accounts underappreciate. Distributional patterns encode not merely which words co-occur but structural information about *why* they co-occur—information about the semantic features that make certain combinations conceptually appropriate. When *cause* appears with forceful impositions, when *foster* appears with gradual developments, when *regime* appears with systematic structures, the distributional evidence carries information about agentivity, facilitation, and system-control respectively. Statistical learning mechanisms in both human cognition and artificial neural networks can extract these structural regularities, enabling generalization beyond surface frequencies.

This reconceptualization of what corpus statistics encode has implications for understanding language acquisition more broadly. If distributional patterns in linguistic input carry information about semantic structure, then statistical learning—whether by children or machines—constitutes a viable mechanism for acquiring abstract semantic knowledge. The puzzle of how learners extract meaning from form becomes less mysterious: the statistical patterns in language use are not arbitrary or divorced from meaning but are consequences of semantic structure interacting with communicative function. Learners who are sensitive to these patterns are therefore indirectly learning about the semantic architecture that generates them.

It is important to acknowledge that this account does not claim language models "understand" in the same way humans do or that statistical learning alone is sufficient for full semantic competence. Models may lack grounding in sensorimotor experience, causal reasoning, and social cognition that inform human meaning. However, the observation that purely distributional training enables approximation of semantic profiles suggests that corpus statistics encode more structural information than previously recognized. The success of statistical models provides evidence for the profile-based account: distributional patterns are consequences of semantic structure, making that structure partially recoverable through statistical learning.

Theoretical Integration

The proposed framework advocates for integration rather than paradigm replacement. It maintains that both usage-based and formal semantic traditions contribute essential insights, but they operate at different levels of explanation and different timescales. At the diachronic level, repeated usage shapes lexical representations through mechanisms well-documented in historical linguistics and grammaticalization research (Bybee, 2010; Traugott & Dasher, 2002). Pragmatic inferences can become conventionalized, syntactic patterns can be reanalyzed as semantic features, and frequency effects can strengthen or weaken particular aspects of meaning. Over centuries, the cumulative effect of usage undoubtedly molds the semantic profiles we observe synchronically.

However, at the synchronic level—the level relevant to explaining observable collocational patterns in contemporary corpora—semantic profiles function as relatively stable constraints. These profiles are products of historical development but, once established, they act as organizing principles that shape rather than merely reflect usage. The collocational patterns corpus linguists discover represent the systematic interaction between these stable semantic features and discourse pragmatics. Understanding this interaction requires analyzing the semantic architecture, not merely documenting its distributional consequences.

This integrated perspective resolves apparent tensions in the literature. Usage-based scholars are correct that meaning is shaped by use over long timescales, while formal semanticists are correct that synchronic patterns are best explained by structured representations. Profile-based accounts and prosody-based accounts need not be mutually exclusive: profiles can be historically emergent from usage while synchronically functioning as constraints on usage. The directionality question receives a nuanced answer: usage shapes meaning diachronically, but meaning shapes usage synchronically.

Linguistic Theory

For linguistic theory, this reconceptualization underscores the necessity of integrating robust lexical-semantic analysis with corpus methodologies. As Hunston (2007, p. 53) cautions, "corpus data are evidence of meaning rather than meaning itself." Corpora are unparalleled descriptive tools revealing hidden patterns, but those patterns demand explanation. A theory that stops at identifying a "negative prosody" provides a label, not an explanation. Explanatory power comes from linking distribution to semantic architecture. This approach also resolves tension between the stability of lexical meaning and fluidity of use by positing a stable core that governs, but does not rigidly determine, contextual deployment.

The framework developed here suggests productive directions for future research. First, explicit formalization of semantic profiles using feature geometry or frame semantic representations could make predictions more precise and testable. Rather than informal descriptions of "agentive" or "facilitative" meanings, researchers could specify feature bundles (e.g., [+DIRECT, +FORCE, +CHANGE] for *cause*) that predict both prototypical uses and acceptable extensions. Second, cross-linguistic investigation could test whether similar profile-pattern relationships hold in typologically diverse languages. If the agentive profile of causatives universally attracts them to harm contexts, this would provide strong evidence for the account. If the pattern varies significantly, it would suggest greater role for language-specific conventionalization.

Third, experimental psycholinguistic research could directly test whether speakers mentally represent semantic profiles as this account claims. Priming studies could examine whether activating profile features (e.g., priming force-dynamic concepts) facilitates processing of appropriate verbs. Acceptability judgment studies could test whether gradient judgments align with profile-context fit predictions. Such research would establish the psychological reality of profiles beyond their explanatory utility.

Lexicography

For lexicography, the implications are practical and actionable. Dictionaries that note collocational tendencies of words like *cause* or *regime* provide valuable usage guidance. However, this analysis suggests that definitions should strive to capture underlying semantic profiles that motivate these tendencies. For instance, defining *cause* with explicit reference to its direct agentivity, rather than merely noting its frequent use with "something bad," provides a more generative and accurate account. It explains both common negative uses and marked positive ones, moving the entry from correlation description to causal explanation.

A revised entry for *cause* might read: "To bring about or produce (an effect or result) through direct, forceful action; implies a salient agent or potent force directly effecting change. Note: While evaluatively neutral, *cause* frequently appears with negative outcomes in discourse, as harmful events are often conceptualized as requiring forceful agents. Compare *lead to* (less direct causation), *foster* (facilitative causation)."

Similarly, defining *regime* to emphasize its system-control semantics, with a usage note about frequency in political contexts, would better serve dictionary users than simply labeling it as "often pejorative." This approach empowers users to understand not just what words tend to mean but why they pattern as they do, providing generative knowledge applicable to novel contexts.

Corpus-informed dictionaries could benefit from distinguishing three types of information: (1) core semantic profile features, (2) statistically frequent collocations, and (3) pragmatic associations arising from their interaction. This tripartite structure would clarify that frequency patterns (2) arise from the interaction of stable meaning (1) with discourse priorities, resulting in pragmatic tendencies (3) that are strong but not absolute. Such clarity would prevent users from mistaking distributional skews for semantic rules.

Natural Language Processing

For natural language processing and computational linguistics, the directionality debate has direct practical consequences. Systems for sentiment analysis, machine translation, and text generation that rely on simplistic polarity lexicons often fail because they mistake pragmatic skew for semantic rule. A sentiment analyzer trained on collocational frequency of *cause* might incorrectly tag "cause for celebration" as negative. A more robust system would incorporate features representing semantic profiles (e.g., [±AGENTIVE], [±INTENSIVE], [±FACILITATIVE]).

Understanding that *cause* has an agentive profile allows a system to better interpret why "cause economic growth" sounds more forceful and less natural than "foster economic growth." The difference is not that *cause* is "negative" while *foster* is "positive," but that forceful direct causation (agentive profile) is pragmatically less felicitous for describing gradual economic processes than facilitative causation. NLP systems that model these semantic distinctions achieve more nuanced, human-like language understanding (Mohammad & Turney, 2013).

Current large language models implicitly learn approximations of semantic profiles from distributional training, as their successful generalization demonstrates. However, explicitly modeling profiles could enhance performance and interpretability. Architectures that learn structured semantic representations—perhaps through techniques like probing classifiers that identify profile features in learned embeddings—could achieve better zero-shot generalization and more robust performance on edge cases. Research in this direction would benefit from linguistically informed feature sets derived from frameworks like Dowty's (1991) proto-roles or Pustejovsky's (1995) qualia structure.

Furthermore, evaluation metrics for language models should test profile-based generalization, not merely perplexity on typical uses. A model that generates only "foster growth" and "cause problems" has learned distributional frequencies; one that appropriately generates "foster resentment" and "cause celebration" has learned semantic structure. Benchmarks emphasizing creative but semantically appropriate uses would better assess whether models have acquired human-like semantic competence.

Language Pedagogy

For language pedagogy, this perspective argues for teaching meaning in depth rather than presenting students with lists of words carrying "bad" or "good" associations. Instructors can explain core semantic features that make particular associations likely. Teaching that *commit* involves binding choice helps learners understand its range from *commit a crime* to *commit to a partner*. This deeper understanding is more transferable and empowering than memorizing collocational lists. It provides learners with generative knowledge that extends to novel contexts rather than rote patterns that may fail in unfamiliar situations.

Consider pedagogical approaches to *cause* and *foster*. Traditional instruction might present them as near-synonyms differing primarily in connotation: "*cause* is used with bad things, *foster* with good things." This approach would confuse learners when they encounter "cause excitement" or "foster dependence." A profile-based approach instead explains: "*cause* implies direct, forceful bringing about; *foster* implies creating conditions that enable gradual development. In English discourse, we more often frame harmful events as forcefully imposed and beneficial developments as gradually nurtured, which is why you'll see these patterns—but the words themselves are evaluatively neutral." This explanation prepares learners for both prototypical and creative uses.

Language textbooks could benefit from exercises that help learners internalize semantic profiles. Rather than gap-fill exercises focusing on correct collocations, activities could ask learners to explain *why* certain combinations sound natural or marked: "Why does 'peace set in' sound better than 'joy set in'? Consider what *set in* means about how a state begins." Such metalinguistic reflection develops deeper semantic understanding that supports flexible, native-like use.

Conclusion

This paper has argued for a significant reconceptualization in how linguists understand the phenomenon often labeled "semantic prosody." Through detailed analysis of *cause*, *utterly*, *commit*, *set in*, *foster*, and *regime*, it has demonstrated that their striking collocational patterns are better understood as consequences rather than causes of their connotative meanings. The direction of explanation flows more powerfully from stable lexical-semantic profiles to observed usage patterns, mediated by discourse pragmatics, rather than from usage to meaning as the strong prosodic account suggests.

The negative skew of *cause* derives from its agentive profile interacting with the types of events we most frequently report; the negative tendency of *utterly* stems from its intensive meaning meeting conventions of evaluative discourse; the association of *commit* with transgression arises from its volitional-binding core attracting it to contexts of weighty, deliberate acts; the link between *set in* and undesirability is rooted in its semantics of gradual onset matching the temporal structure of commonly discussed negative processes; the positive lean of *foster* emerges from its facilitative nature aligning with how we describe cultivated outcomes; and the negative aura of *regime* is a consequence of its system-control profile intersecting with dominant political discourse topics.

Corpus-attested counterexamples for each word undermine claims of inherent, fixed evaluative prosodies and confirm that core semantic profiles remain constant, generative sources of meaning across diverse contexts. Moving beyond semantic prosody therefore means moving beyond description to explanation. It requires recognizing that words carry inherent semantic shapes—structured sets of features defining prototypical characteristics—that selectively attract them to particular corners of conceptual and discursive space. The connotative patterns observed in corpora are results of this attraction, filtered through the preoccupations and priorities of human discourse.

This lexical-semantic-first model does not deny the influence of usage on language change over long timescales, but it asserts the explanatory priority of encoded meaning for understanding stable, systematic connotative behaviors of the core lexicon at any given synchronic moment. By formalizing this insight—shifting from a model of "connotative absorption from context" to one of "profile-constrained deployment according to inherent features"—we can reintegrate sophisticated understanding of connotation into the heart of linguistic theory.

The proposed integrated model acknowledges that lexical representations are shaped by usage diachronically while maintaining that synchronic distributional patterns are better explained by stable semantic architecture interacting with discourse pragmatics. This reconceptualization bridges rich corpus data with explanatory depth from formal and cognitive semantics, offering a more complete and principled account of how words mean and how that meaning shapes the way we deploy them to understand, describe, and evaluate our world.

The parallel implications for understanding statistical language models add contemporary urgency to these theoretical debates. The success of systems like GPT-4 in acquiring human-like semantic competence from purely distributional training is not mysterious once we recognize that corpus statistics encode information about semantic structure. Collocational patterns are not arbitrary associations but consequences of semantic profiles interacting with discourse pragmatics. Statistical learning mechanisms, whether in biological or artificial neural networks, can extract these structural regularities precisely because they are systematically encoded in distributional patterns. This suggests that the opposition between "mere statistics" and "real understanding" may be overdrawn: statistics derived from meaningful language use carry information about the semantic structures that generated them, making those structures partially recoverable through statistical learning.

Future research should pursue experimental validation of proposed semantic profiles through psycholinguistic methods, cross-linguistic comparison to test the universality of profile-pattern relationships, and computational modeling that operationalizes profile features for NLP applications. Investigating whether languages with different discourse conventions show similar or different mappings between profiles and distributions would illuminate the relative contributions of universal semantic structure versus language-specific pragmatic conventionalization. Developing explicit formal representations of semantic profiles would enable more precise predictions and computational implementations.

By recognizing that semantic prosody research has discovered not evaluative absorption but the distributional fingerprints of semantic architecture, we achieve theoretical unification: corpus patterns become explicable through lexical semantics, usage-based and formal approaches are integrated across timescales, counterexamples are seamlessly accommodated, and the success of statistical language models in approximating human semantic competence becomes understandable. The path forward requires continued dialogue between corpus linguistics, formal semantics, psycholinguistics, and computational approaches—each contributing essential insights to our understanding of how meaning shapes, and is shaped by, the statistical fabric of language use.

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